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Universal Machine Learning Interatomic Potentials on HPC

A joint webinar from ENCCS and Sweden AI Factory
Wednesday, 30 September 2026, 10:00 CEST

Hands-on lesson: enccs.github.io/mlip-hands-on

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scan for the lesson

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- ENCCS, the EuroCC National Competence Centre Sweden: free training and expert support in HPC, AI and data science, with access support for EuroHPC systems such as LUMI and Leonardo · enccs.se
- Sweden AI Factory, a node in the EuroHPC AI Factories network: compute, AI expertise and onboarding, with a new AI-optimised supercomputer arriving in Linköping in 2026/2027; materials science is among its focus areas · swedenaifactory.se

Questions during the webinar: please use the Zoom Q&A. Contact: training@enccs.se

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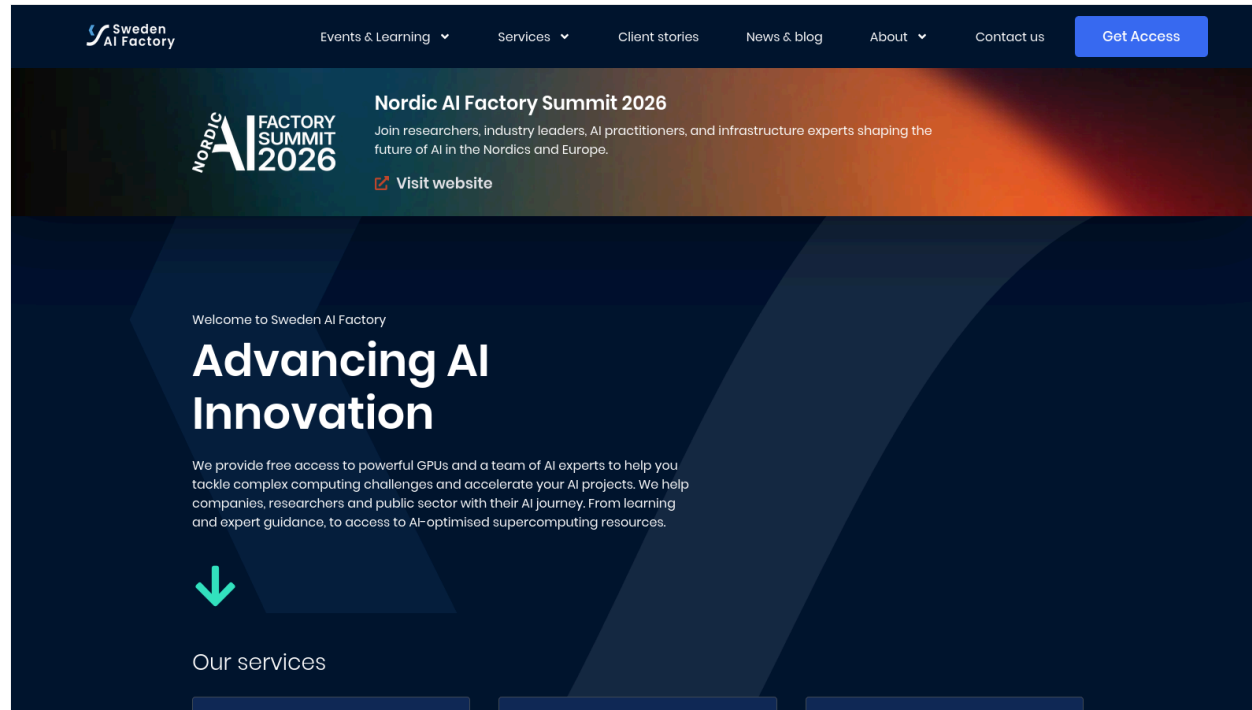


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The screenshot shows the Sweden AI Factory website. The top navigation bar includes links for 'Events & Learning', 'Services', 'Client stories', 'News & blog', 'About', 'Contact us', and a 'Get Access' button. The main banner features the 'Nordic AI Factory Summit 2026' logo and text: 'Nordic AI Factory Summit 2026. Join researchers, industry leaders, AI practitioners, and infrastructure experts shaping the future of AI in the Nordics and Europe. Visit website'. Below this, a section titled 'Welcome to Sweden AI Factory' is followed by the heading 'Advancing AI Innovation'. A paragraph describes the service: 'We provide free access to powerful GPUs and a team of AI experts to help you tackle complex computing challenges and accelerate your AI projects. We help companies, researchers and public sector with their AI journey. From learning and expert guidance, to access to AI-optimised supercomputing resources.' A green arrow points down to the 'Our services' section.

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What we will cover

1. The simulation bottleneck: accuracy vs cost, Jacob's ladder, what an MLIP is, where MLIPs sit in AI for materials, xTB next door
2. Foundation models for atoms: data, architectures, UMA and OMol25, our Orb vs UMA test on LUMI, fast universal models (Orb and friends)
3. Pre-train, then fine-tune: data efficiency, where zero-shot falls short, a practical recipe
4. Running them fast: GPU-native engines, batching, Leonardo, LUMI and Arrhenius
5. Choosing and trusting a model: benchmarks, leaderboards, transition states (NEB), rules of thumb
6. Outlook, then hands-on: long-range physics, ML inside DFT, generative models, agents; the companion lesson

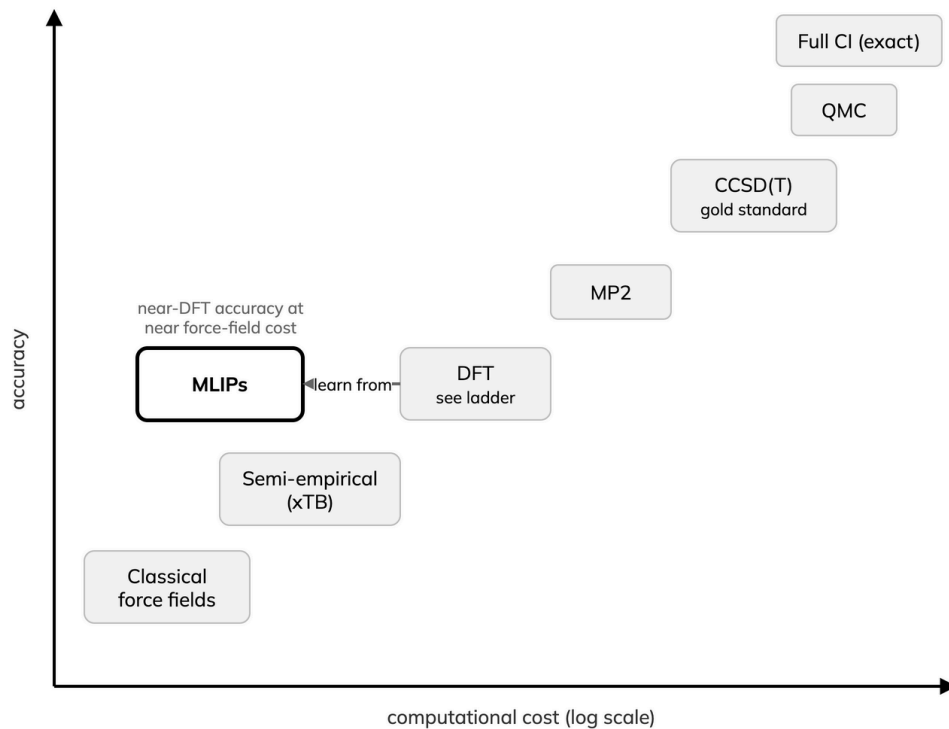
Then Q&A

A few words that could be relevant here

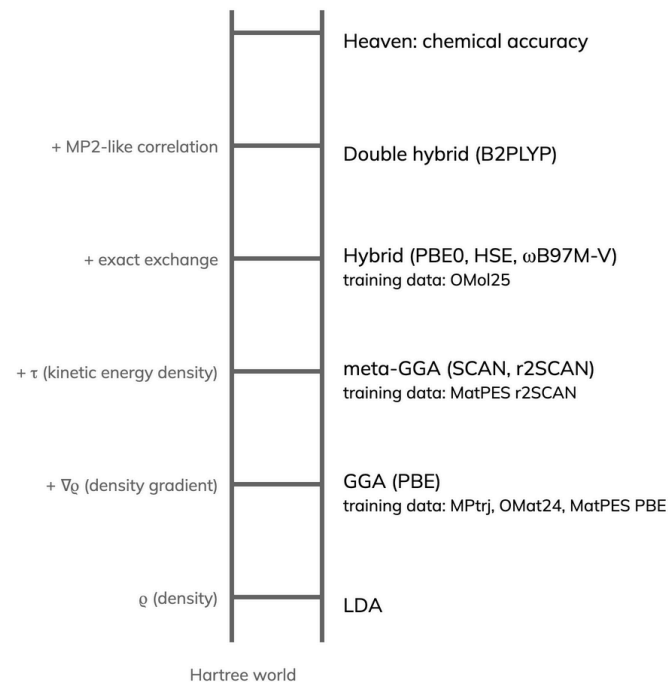
Term	In plain language
Universal / foundation model	One model pre-trained on huge data, then reused for many problems
Zero-shot	Used straight out of the box, with no extra training
Fine-tuning	A little extra training to specialise the model for your system
Graph neural network	Atoms exchange information with their neighbours, in rounds
Equivariant	The prediction rotates correctly when you rotate the molecule
Transformer	The model family behind large language models such as ChatGPT
meV/atom	The accuracy unit: energy error per atom against the first-principles reference; smaller is better
DFT (GGA, hybrid)	First-principles method that generates the training data; the rung (GGA, meta-GGA, hybrid) sets its accuracy
Dispersion (D3)	The weak van der Waals attraction; missing in PBE-trained models, added as a correction
F1, κ SRME, CPS	Leaderboard scores: stable-crystal classification (higher is better), thermal-conductivity error (lower is better), and CPS, their combined score
NEB	Nudged elastic band: finds the path and energy barrier between two atomic arrangements

1 · The simulation bottleneck

The accuracy versus cost trade-off



Jacob's ladder of DFT



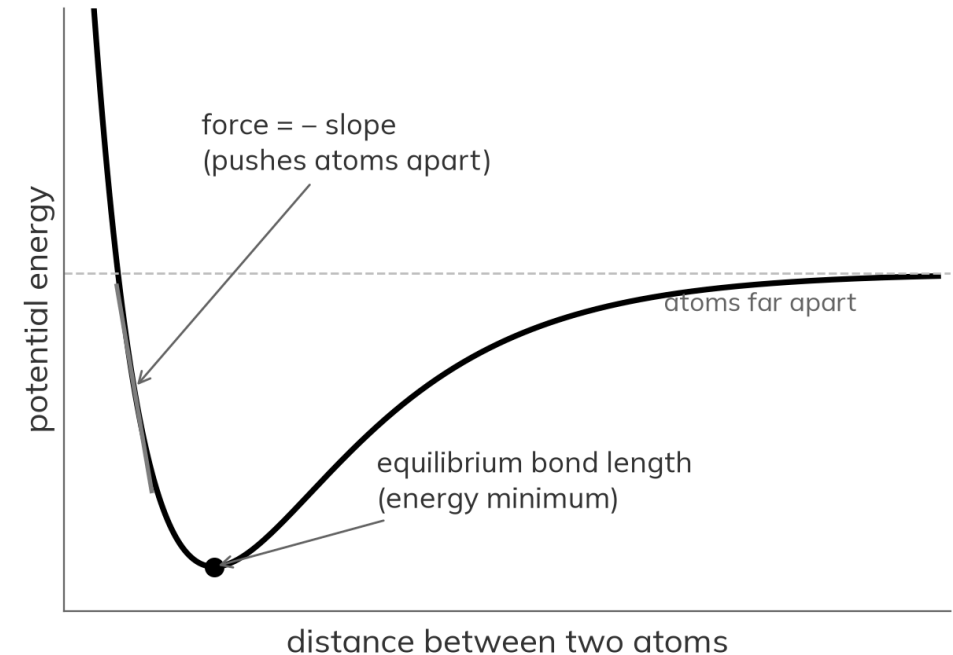
MLIPs reuse one rung's accuracy at a fraction of its cost; they inherit that rung's accuracy, not more.

What an interatomic potential is

A potential gives the system's energy as a function of where the atoms are, the potential energy surface (PES). Its slope gives the forces that drive molecular dynamics and structure relaxation.

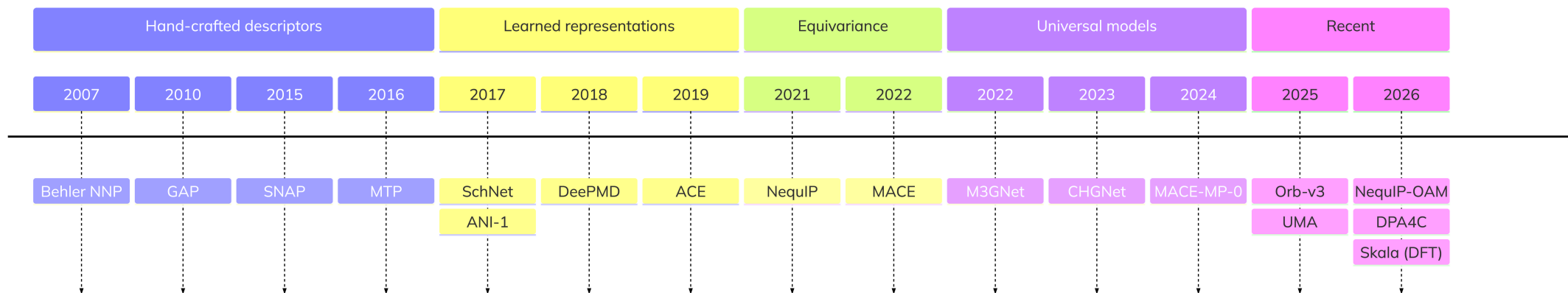
- Classical force fields fix a simple hand-crafted form (springs, Lennard-Jones, Coulomb). Fast, but limited accuracy and transferability.
- First-principles (quantum-mechanical) methods such as DFT compute the energy from electronic structure. Accurate, but cost grows steeply with system size.

The dream: DFT-quality energies and forces at force-field-like cost.



Energy of two atoms versus their distance (illustrative Morse curve).

The machine-learning idea



Behler 2007 [1, 2] · GAP 2010 [3, 4] · SNAP 2015 [5, 6] · MTP 2016 [7, 8] · ANI-1 2017 [9, 10] · ANI-1ccx, CCSD(T) data [11] · SchNet [12, 13] · DeePMD 2018 [14, 15] · ACE 2019 [16, 17] · NequIP [18, 19] · MACE [20, 21] · M3GNet [22, 23] · CHGNet [24, 25] · MACE-MP-0 [26, 27] · UMA [28, 29] · Orb-v3 [30, 31] · NequIP-OAM [32, 19] · DPA4C [33, 15] · Skala (learned DFT) [34, 35]

Where MLIPs sit in AI for materials

Six families of methods, often confused. They differ in what the model learns or does.

📈 Electronic structure

learns the quantum mechanics itself

- Neural wavefunctions (FermiNet, PauliNet)
- Learned XC functionals (DM21; Skala)
- ML Hamiltonians and tight-binding (DeepH)

⊗ Interatomic potentials

learns energies and forces from DFT data

- Universal models (MACE-MP, UMA, Orb, MatterSim, NequIP-OAM)
- Fine-tune a foundation model on your own data

✦ Generative models

proposes new structures on demand

- Diffusion for crystals (CDVAE, MatterGen)
- Language models write crystal files (CrystaLLM)
- Organic crystal prediction (Clari, 2026)

🔗 Structure to property

maps a structure straight to a property

- Descriptors (SOAP, matminer)
- Crystal graph networks (CGCNN, ALIGNN)
- Multi-task models (MatterSim-MT, 2026)

🤖 LLMs and agents

reads, plans and runs the tools

- Chemistry agents (ChemCrow, Coscientist)
- Simulation agents (MDCrow, El Agente, AtomisticSkills)
- LLM with an MLIP encoder (MatterChat)

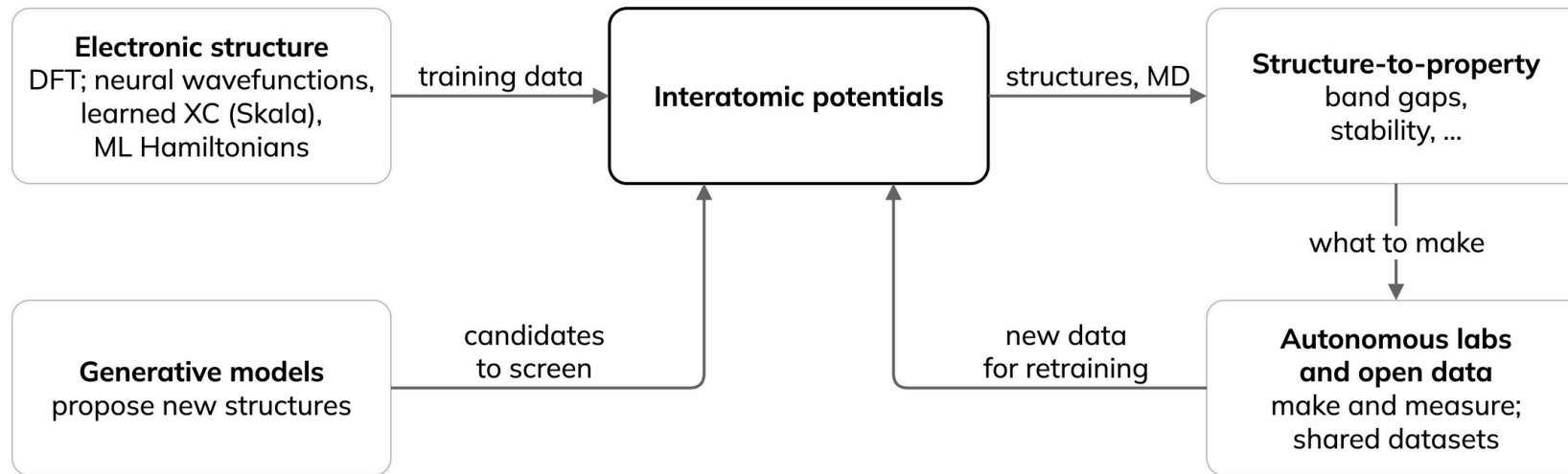
🧪 Autonomous labs, open data

closes the loop: predict, make, measure

- Self-driving labs (A-Lab, robotic chemist)
- Open data (OMat24, Alexandria, LeMaterial)
- Predicted stable is not always synthesisable [72, 73]

Open data trains potentials and property models; generators propose, agents orchestrate, labs test and feed new data back.

How the families connect



LLM agents
plan and run any step:
DFT, MLIP relaxations
and MD, generators,
property models,
lab robots,
data analysis

- MLIPs sit in the middle: first-principles data in, structures and dynamics out
- Generators propose, MLIPs screen, labs make; new data feeds retraining
- LLM agents can run any step, from one DFT or MLIP calculation to the whole loop

Next door: fast quantum chemistry (xTB)

The Grimme group (Bonn) builds fast, physics-based tools that need no training data.

⚡ The xTB family

- g-xTB: near hybrid-DFT quality, elements H to Lr [36]
- CREST: xTB samples, an MLIP (MACE, UMA) refines [37]
- Any element, charges and spin, out of the box

√ D3/D4 dispersion

- Adds the van der Waals that PBE-trained MLIPs miss
- Now on the GPU inside TorchSim and with Orb
- Faster D3 for large cells (2026)

⌘ xTB meets ML

- NN-xTB: a network tunes the xTB Hamiltonian [38]
- xTB plus ML for MOF band gaps (Walsh, 2026) [39]
- dxtb: differentiable xTB in PyTorch [40]

Use xTB when

- Chemistry absent from MLIP training data
- Unusual elements, charge or spin states
- Electronic properties: charges, orbitals, gaps

Use an MLIP when

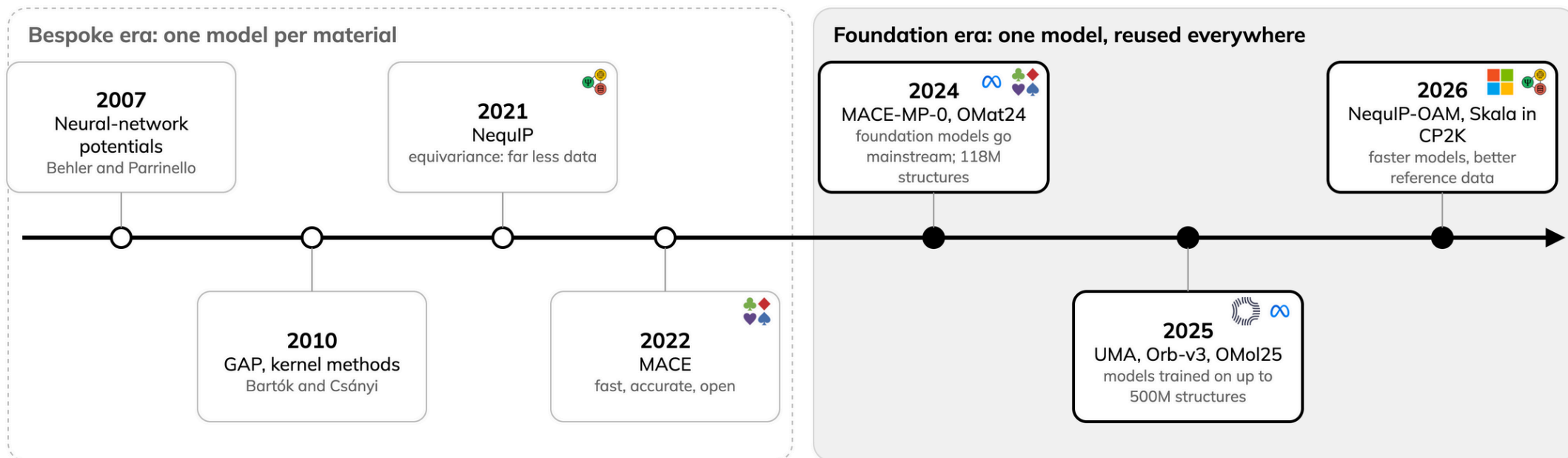
- System within the training distribution
- Near-DFT accuracy in that domain
- Large systems, long MD: linear scaling, GPUs

Combine them

- Conformer search with xTB (CREST), energies refined with the MLIP
- xTB as a cross-check outside the MLIP's domain
- D3/D4 dispersion added to either

2 · Foundation models for atoms

From bespoke potentials to foundation models



- Classical force field: one fixed, hand-tuned formula
- System-specific MLIP: trained for one material, refitted for the next
- Universal (foundation) MLIP: pre-trained across the periodic table, reused everywhere

The data explosion

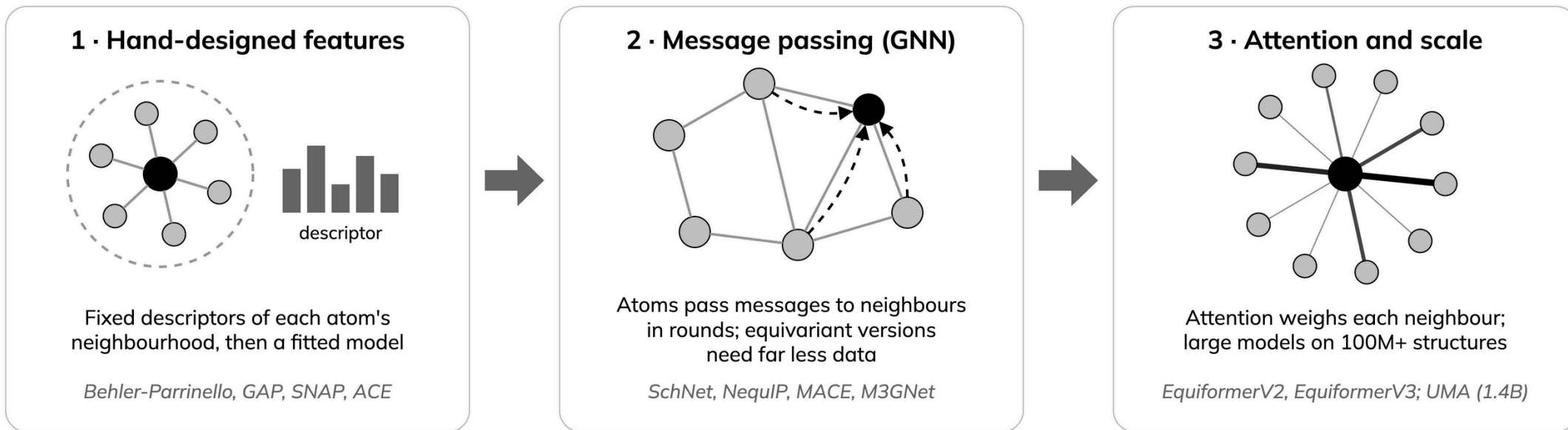
Foundation MLIP training sets have grown more than a hundredfold in a few years:

Dataset	Scale	Reference level
MPtrj	~1.58M configurations, ~146k Materials Project compounds, 89 elements	PBE(+U)
MatterSim	~17M configurations (active learning)	PBE(+U)
GNoME	~89M structures (active learning, not public)	PBE(+U)
OMat24	~118M inorganic structures	PBE+U
OMol25	>100M calculations, ~83M molecular systems, 83 elements	ω B97M-V/def2-TZVPD
UMA training	~500M unique 3D structures	mixed

MPtrj [\[24\]](#); OMat24 [\[41\]](#); OMol25 [\[42\]](#); UMA [\[28\]](#)

How the models are built

From features designed by hand to features the model learns from data:



- Equivariant: rotate the structure and the forces rotate with it; NequIP needed up to about 1000x less data [\[18\]](#)
- By 2026 simpler designs such as Orb-v3 compete: data and scale matter most [\[30\]](#)

Examples: GAP [\[3\]](#), ACE [\[16\]](#), MACE [\[20\]](#), EquiformerV2 [\[43\]](#), UMA [\[28\]](#)

A field of foundation models

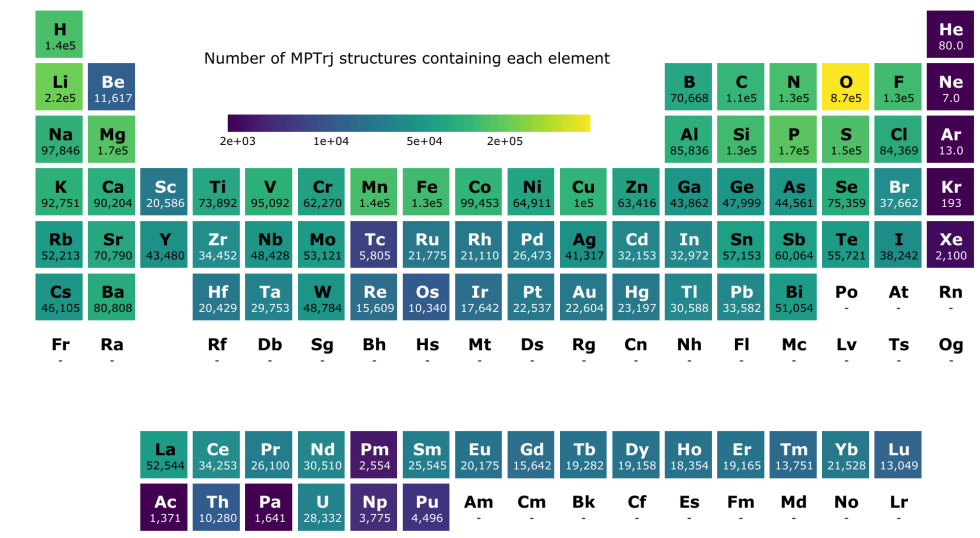
Model family	From	Architecture	Params	Training data
CHGNet (2023) [24, 25]	UC Berkeley	GNN with magnetic moments	0.4M	MPtrj
MACE-MP-0 (2024) [26, 27]	Cambridge and others	equivariant message passing (ACE)	4.7M	MPtrj; newer heads add more
SevenNet-0, now Omni [88, 46]	SNU	NequIP-style equivariant GNN	0.8M → 55M	MPtrj → 243M structures
Orb-v3 [30, 31]	Orbital Materials	non-equivariant graph network	26M	OMat24, or MPtrj + Alexandria
eqV2 → eSEN → UMA [83, 28, 29]	Meta FAIR	equivariant transformer → eSEN with mixture of experts	31–153M; UMA to 1.4B (50M active)	OMat24 → ~500M
DPA-3 → DPA-4, DPA4C [33, 15]	AlSI Beijing, DP Technology	line-graph GNN → SO(3)-equivariant	0.03–25M	OpenLAM (163M)
NequIP / Allegro-OAM (2026) [32, 19, 89]	Harvard, Cambridge	E(3)-equivariant; Allegro strictly local	0.6–32M	OAM
PET-OAM (2026) [84, 86]	EPFL	transformer, symmetry not enforced	26–730M	OAM
GRACE [85, 87]	ICAMS, Bochum	graph atomic cluster expansion	3.4–42M	MPtrj; OAM

OAM = OMat24 + sAlex + MPtrj. Parameter counts from the papers and Matbench Discovery model files; MatterSim and other fast models are on the next slide.

“Universal” means across the periodic table

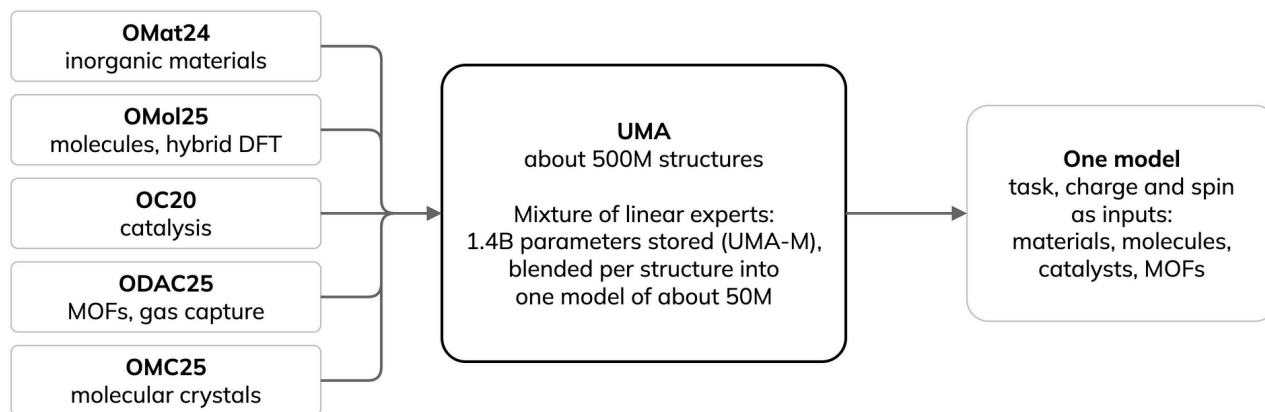
A foundation MLIP covers most of the periodic table out of the box. Coverage reflects the training data: common elements appear in hundreds of thousands of structures, rare ones such as the noble gases in only a handful.

A useful sanity check is whether the model gives physically correct behaviour everywhere, for example repulsion at short interatomic distances for every element.



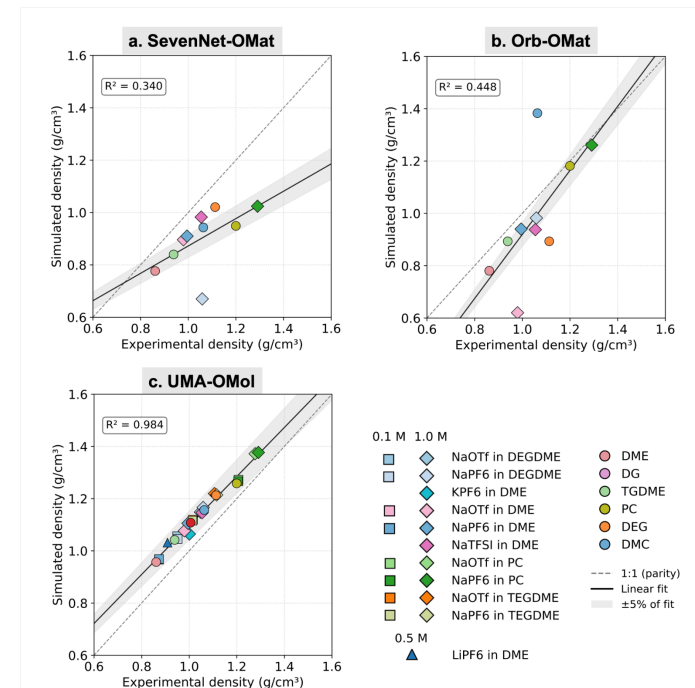
Number of MPTrj structures containing each element. Matbench Discovery [51, 52] (MIT)

Scale arrives: the UMA example



- One model, five datasets (2025); the experts are blended per structure, so it runs like a 50M model
- OMol25 is at hybrid-DFT level (ω B97M-V), a rung above PBE
- It matters: electrolyte densities R^2 0.98, against 0.34 and 0.45 for materials-only models (right)
- Current checkpoints: UMA 1.2 small (patch 1.2.1); uma-s-1 is archived

UMA, Meta FAIR [28, 29]; OMol25 [42]



Electrolyte densities, simulation vs experiment. Kumar et al., [arXiv:2603.20183](https://arxiv.org/abs/2603.20183), Fig. 1 [45] (CC BY 4.0, cropped)

Our LUMI run: one model for crystals and molecules

Orb-v3 and OrbMol against UMA 1.2 small, on one MI250X GCD, float64 ([lesson A6](#)):

Quantity	Orb-v3 / OrbMol	UMA	Reference
Si lattice constant (Å)	5.460	5.461	PBE 5.470, exp. 5.431
Cu lattice constant (Å)	3.613	3.615	PBE 3.629, exp. 3.615
Water ionisation energy (eV)	6.07	12.64	exp. 12.62
O ₂ singlet minus triplet (eV)	1.54	1.50	exp. 0.98
Time per force call, 216 Si atoms (s)	0.06	0.14	

- Crystals: both within 0.5% of PBE, as expected from their training data
- Charge: UMA matches the water ionisation energy; OrbMol gives half, so test the property you need
- Spin: both overestimate the O₂ gap; the OMol25 reference treats that singlet as closed-shell

Fast universal models: Orb and friends (2026)

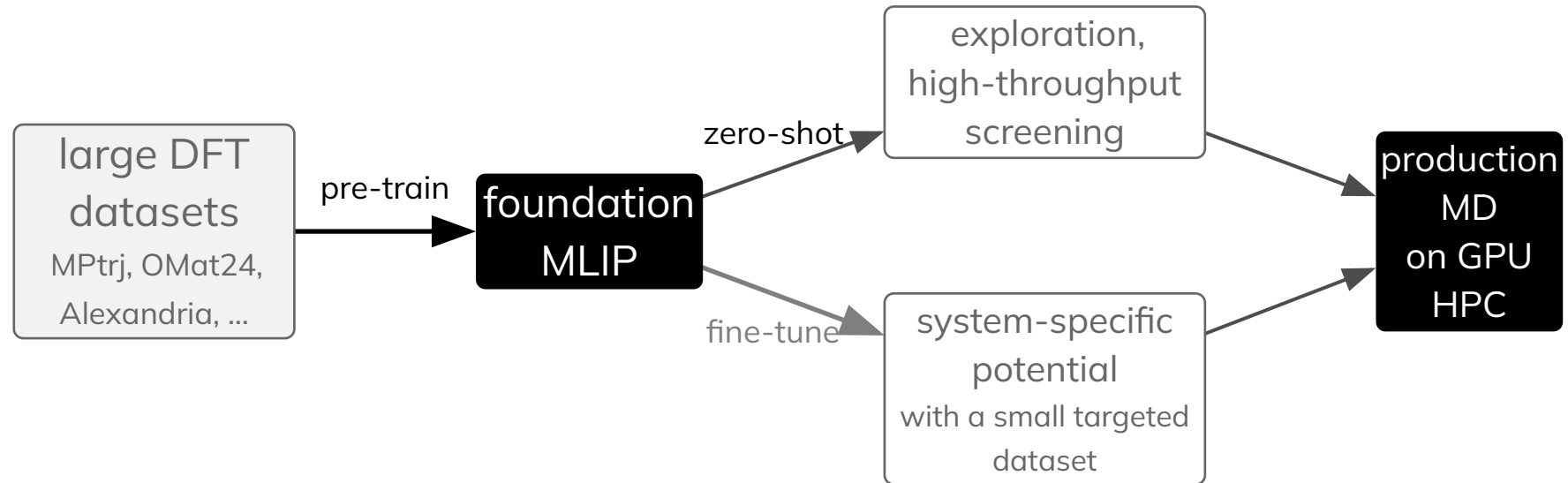
Model	From	Params	F1 ↑	κ SRME ↓	Licence	Why pick it
Orb-v3 [30, 31]	Orbital Materials	26M	0.905	0.21	Apache-2.0	fast; D3 variant for van der Waals
SevenNet-Omni [88, 46]	SNU	55M	0.906	0.19	MIT	D3 built in; LAMMPS and TorchSim
NequIP-OAM-XL [32, 19]	Harvard	32M	0.906	0.13	MIT / CC-BY	also runs on AMD GPUs (LUMI)
MatRIS-10M-OAM [90, 47]	ICT, CAS	10M	0.921	0.22	BSD-3	best accuracy for its size
MatterSim v1 5M [44, 48]	Microsoft	4.5M	0.862	0.57	MIT	small and fast
Nequix [91, 49]	MIT	0.7M	0.751	0.45	MIT / CC-BY	cheapest to run and train
eSEN-30M-OAM [83, 29]	Meta FAIR	30M	0.925	0.17	MIT / gated	very accurate; UMA family
EquiformerV3-OAM [92, 50]	MIT	30M	0.931	0.12	MIT	accuracy leader, slower

- F1: stable-crystal classification, higher is better; κ SRME: thermal-conductivity error, lower is better
- Most GPU speed-ups are NVIDIA-only (Leonardo, Arrhenius); on LUMI pick a pure-PyTorch model such as NequIP or MACE
- Not on the leaderboard: MatGL [23] 4.0.3 (BSD-3), small TensorNet, CHGNet, M3GNet and QET models trained on MatPES; no TorchSim interface, so it runs through ASE

Matbench Discovery [51] leaderboard data, accessed 29 Sept 2026 [52]

3 · Pre-train, then fine-tune

The workflow

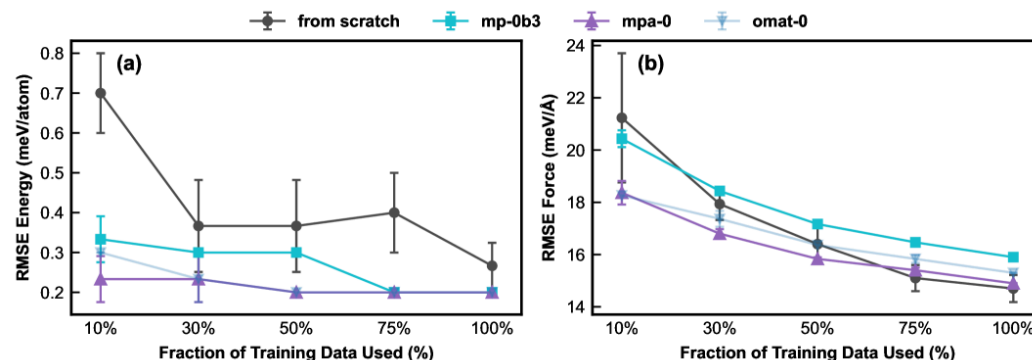


Use the foundation model directly for exploration, or fine-tune it on a small, targeted dataset when you need quantitative accuracy for one system.

Fine-tuning needs less data

A pre-trained model already encodes broad chemistry, so fine-tuning usually needs fewer configurations than training from scratch; in the cases below it matched or beat it.

- Fine-tuned models have lower energy errors at every data fraction, with the biggest gain at 10%; force errors converge with from-scratch as data grows
- High-entropy alloy: 13.8 meV/atom fine-tuned vs 16.4 (MACE) and 24.1 (ACE) from scratch ([53])
- Our LUMI run ([lesson A4](#)): TensorNet, PBE to r2SCAN, 84 Li structures: 60 vs 513 meV/atom from scratch, forces 128 vs 427 meV/Å (one run, one MI250X GCD)

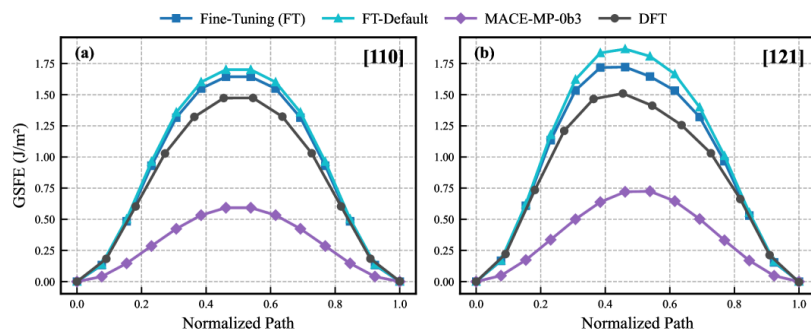


Learning curves, fine-tune vs from scratch [54] (CC0)

Where zero-shot models fall short

Used as-is, foundation models get structures right but can be far off on mechanical and disordered systems; fine-tuning closes most of the gap.

System, model	Property	Zero-shot	Fine-tuned
Mo, MACE-MP-0b3 [54]	C_{11} elastic constant, error	45.9%	2.6%
Mo, MACE-MP-0b3 [54]	stacking-fault energy (below)	far too low	close to DFT
Si, MACE-MP-0b [53]	elastic constants, error	19–53%	0.6–5.2%
High-entropy alloy, MACE-MP-0 [53]	energy error, meV/atom	59–64	13.8
41 models on α -C and α -SiO ₂ [55]	energy error	>100% for some (α -C)	>5x lower, 4 structures of α -SiO ₂



- Elastic constant: how stiff a material is
- Stacking-fault energy: the cost of sliding atomic layers; sets how metals deform

Mo stacking-fault energy: MACE-MP-0b3 (purple), DFT (black), fine-tuned (blue) [54] (CC0)

A practical recipe

► 1 · Start zero-shot

Use the foundation model directly; often good enough for screening and exploration



🔧 2 · Fine-tune

Add a small, targeted dataset when you need numbers for one system



🎯 3 · Select data

Uncertainty-aware sampling adds what the model is least sure about



✅ 4 · Validate

Test the property that matters, not just the energy

🔗 Catastrophic forgetting

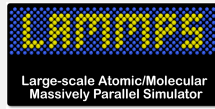
A network trained further on new data can lose accuracy on what it learnt before [\[93\]](#). Fine-tuned universal MLIPs show it too; forgetting-aware fine-tuning limits it [\[94\]](#). Keep the original model for general use.

Recipe [\[54\]](#)

4 · Running them fast: GPU-native engines

The CPU-era stack meets GPU-era models

CPU-era simulation tools



- Built to run one system at a time
- GPU speed-ups first targeted classical force fields
- Excellent for cheap, simple functional forms



GPU-era foundation MLIPs



- Neural networks with millions of parameters
- Want batched inference: many structures per call
- One structure at a time leaves the GPU mostly idle

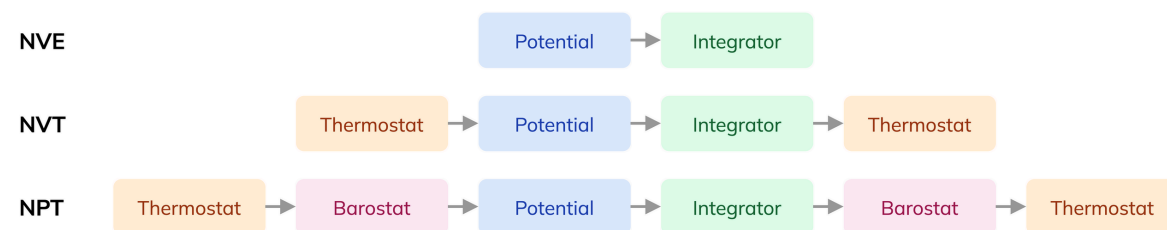
A new generation of engines is built for the way these models compute (next slide).

Logos from the ASE, LAMMPS, GROMACS, PyTorch and JAX projects, used to identify the software.

Engines built for MLIPs

Engine	Stack	Idea
TorchSim [95, 57]	PyTorch	Batched MD, relaxation and Monte Carlo; drives MACE, FairChem/UMA, SevenNet, ORB, MatterSim
kUPS [97, 98]	JAX	Differentiable MD / Monte Carlo / optimisation primitives; runs MACE and UMA
ALCHEMI [99, 101]	PyTorch / Warp	Vendor toolkit; batched and multi-GPU MD and relaxation with MACE, AIMNet2, UMA, MatGL TensorNet

- All open source; all batch many systems on the GPU
- ALCHEMI also supplies common GPU building blocks (neighbour lists, D3, Ewald sums) [\[100, 102\]](#) used by UMA, Orb, PET and TorchSim; NVIDIA GPUs only
- Swappable blocks: the same potential, integrator and thermostat give NVE, NVT or NPT (right)



Engine building blocks, after the kUPS design (CuspAI, 2026).

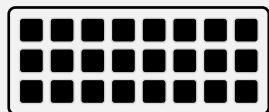
TorchSim (Radical AI, MIT) [\[95, 57, 96\]](#); kUPS (CuspAI, Apache-2.0) [\[97, 98\]](#); NVIDIA ALCHEMI Toolkit (Apache-2.0, v0.2) [\[99, 100, 101\]](#); overview [\[56\]](#)

What batching changes



One system per call

Classical tools and ASE: one structure at a time, hardware mostly idle



Many systems per call

GPU-native engines fill the card with a batch

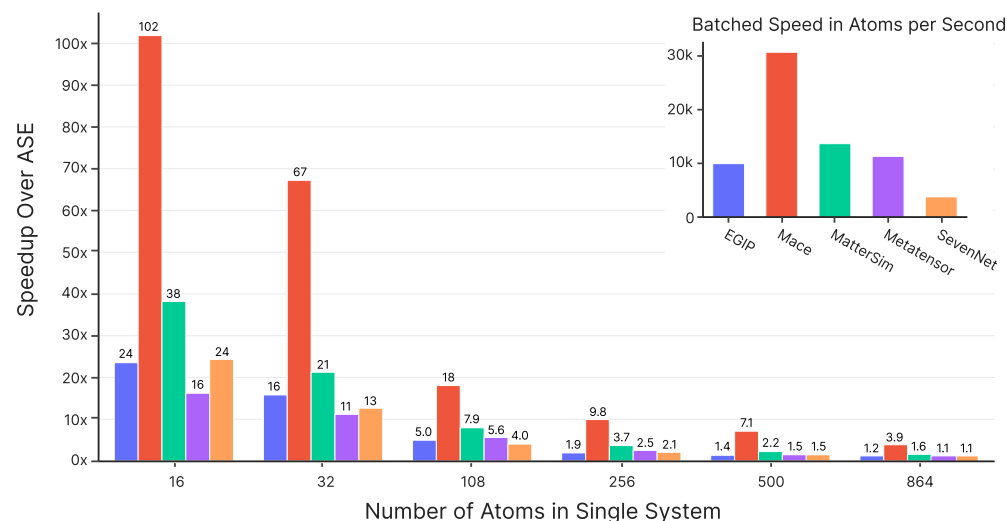
up to ~100x

TorchSim vs ASE, H100; total throughput, not per system [95]

5 to 7x

our A100 runs, 64 to 128 relaxations ([lesson A1](#))

Simulation Throughput of Batched TorchSim vs Serial ASE



Throughput vs ASE (single H100, batched). TorchSim [57] (MIT).

This makes relaxing and screening large candidate sets practical.

On EuroHPC systems

System	GPUs	Practical notes for MLIPs
Leonardo	NVIDIA A100	• Booster nodes with four A100 GPUs each • CUDA-native: TorchSim, cuEquivariance and ALCHEMI work directly • Our runs: batched relaxation, 5 to 7x over serial (A1)
LUMI	AMD MI250X	• LUMI-G nodes with four MI250X, each two GCDs (eight GPUs to software) • ROCm PyTorch runs MACE, NequIP and MatGL (float64); no CUDA-only kernels • Our runs: MACE, MatGL, fine-tuning and NEB on one GCD (A1, A3 to A5)
Arrhenius [71]	NVIDIA GH200	• 382 nodes with four GH200 each; Linköping, inaugurated Sept 2026 • CUDA-native, like Leonardo; ALCHEMI and LAMMPS with MACE run directly • Our runs: 8 MACE MD runs, 18.7 s as one batch vs 88 to 100 s as 8 LAMMPS processes (B7); 4 GPUs 2.4x faster than 1 (B8)

- ENCCS and Sweden AI Factory help with access and the software stack; Sweden AI Factory's own AI-optimised system in Linköping follows in 2026/2027
- NequIP and Allegro foundation models run MD on both GPU types (LAMMPS ML-IAP/Kokkos): 102.5M atoms on 256 GPUs; NequIP-OAM-XL matches eSEN-30M-OAM at about 10x the speed [\[32\]](#)
- DPA4C (DeePMD-kit 3.2.0) approaches MACE-OMat accuracy at about 100x the throughput; 2.048 billion atoms on 1,024 V100s [\[33\]](#)

LUMI: docs.lumi-supercomputer.eu [\[58\]](#). TorchSim added AMD/ROCm support in v0.4.2, verified on consumer cards.

5 - Choosing and trusting a model

One error number is not enough

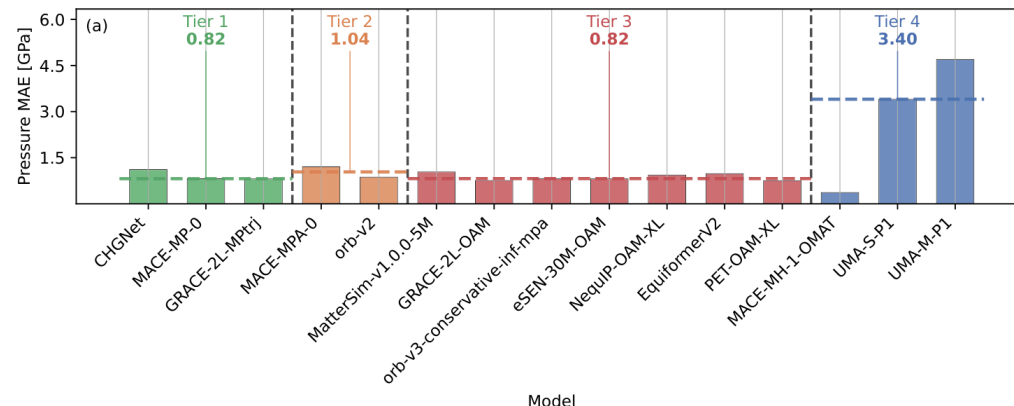
A low force error is not enough: check MD stability, speed, memory and your property [103].

Dyna-Mat: 15 models, finite-temperature MD, tier medians [59]

Tier (training data)	Force RMSE, eV/Å	Pressure MAE, GPa
1 (MPtrj)	0.166	0.82
2 (+ Alexandria)	0.106	1.04
3 (OMat24-based)	0.063	0.82
4 (multi-dataset)	0.054	3.40

Molecules: 15 models, MD of 2,661-atom water on one H100 [60, 104]

Model	Energy MAE, kcal/mol	MD steps/s
UMA-m-1.1	0.53	0.16
UMA-s-1.1	0.61	3.65
MACE-OFF23 (L)	1.73	1.43
AIMNet2	2.55	33.7



Pressure error per model. Gawkowski et al., Fig. 4(a) [59] (CC BY-SA 4.0, cropped)

- Lowest force error, worst pressure: tier 4, driven by the UMA models on one alloy
- Accuracy tracks model size and data, but slower is not always more accurate
- Dyna-Mat is now the Matbench Discovery MD task [105]

Matbench Discovery

- The main public leaderboard for crystal stability: does a predicted crystal hold together or decompose? [\[51\]](#)
- F1, 0 to 1: how reliably the model flags stable crystals

Precision: predicted-stable ones that are stable
Recall: stable ones the model finds

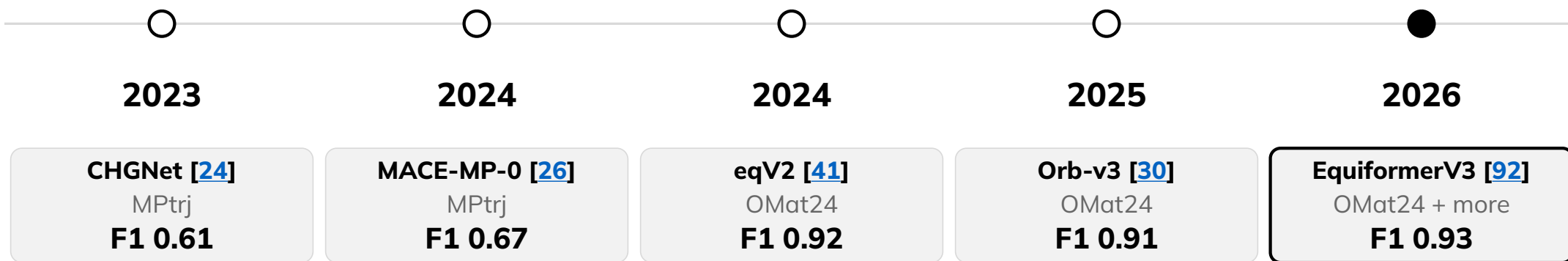
$$F_1 = \frac{2PR}{P+R}$$

- Ranked by CPS, a combined score: 50% F1, 40% thermal-conductivity error (κ SRME), 10% structure error (RMSD)
- Compliant tier: MPtrj-only training, for a fair comparison
- Since July 2026 also an MD task [\[105\]](#)

Model	F1 ↑	DAF ↑	Prec ↑	Acc ↑	TPR ↑	TNR ↑	MAE ↓	RMSE ↓	R ² ↑
eqV2 S DeNS	0.798	4.36	0.748	0.927	0.855	0.941	0.035	0.084	0.785
DPA3-v1-MPtrj	0.757	4.13	0.709	0.911	0.812	0.931	0.04	0.082	0.793
ORB MPtrj	0.755	4.19	0.719	0.911	0.795	0.936	0.043	0.09	0.752
SevenNet-l3i5	0.751	4.11	0.706	0.909	0.803	0.931	0.042	0.086	0.773
SevenNet-0	0.719	3.8	0.653	0.893	0.8	0.912	0.046	0.09	0.75
GRACE-2L (r6)	0.687	3.71	0.637	0.884	0.746	0.912	0.05	0.092	0.74
MACE-MP-0	0.668	3.4	0.583	0.867	0.781	0.885	0.055	0.099	0.698
CHGNet	0.612	3.04	0.521	0.839	0.74	0.859	0.061	0.1	0.69
M3GNet	0.576	2.65	0.454	0.802	0.788	0.804	0.072	0.115	0.588
ALIGNN	0.565	2.92	0.501	0.829	0.649	0.866	0.092	0.154	0.274
MEGNet	0.513	2.7	0.463	0.813	0.574	0.862	0.128	0.204	-0.277
CGCNN+P	0.51	2.4	0.411	0.779	0.67	0.801	0.108	0.178	0.027
CGCNN	0.51	2.63	0.451	0.807	0.587	0.852	0.135	0.229	-0.624
Wrenformer	0.479	2.13	0.365	0.741	0.693	0.751	0.105	0.182	-0.02
BOWSR	0.437	1.84	0.315	0.702	0.711	0.68	0.114	0.164	0.142
Voronoi RF	0.344	1.51	0.259	0.665	0.511	0.697	0.141	0.206	-0.316
Dummy	0.194	1	0.168	0.68	0.231	0.77	0.12	0.181	0

Compliant (MPtrj-only) models, 2025 paper snapshot; best today is F1 $\approx$ 0.86 (EquiformerV3). Matbench Discovery [\[52\]](#). PBE references; not a Materials Project endorsement.

The leaderboard is a moving target



- In three years F1 rose from about 0.6 to 0.93, mainly from more and broader training data
- With training fixed to MPtrj (compliant tier), the best went from 0.82 (2024) to about 0.86 today
- The top models are now within a few hundredths: whatever ranks first will be overtaken within months, so choose by your task, speed and licence

matbench-discovery.materialsproject.org [51], accessed 2026 [52]

Benchmarks beyond a single score

✂ MLIP Arena [61, 106]

Four test groups beyond error metrics:

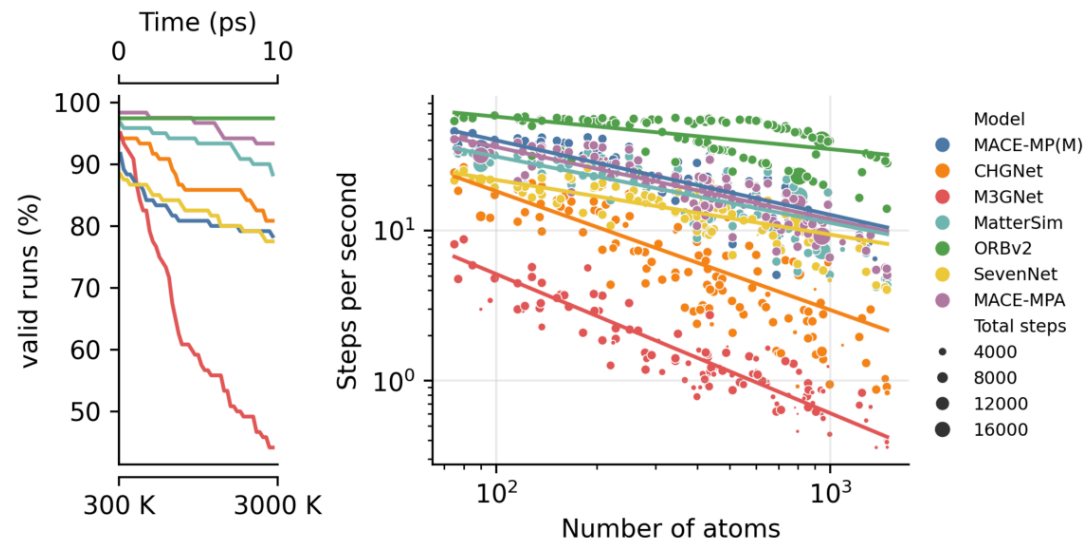
- asymptotic behaviour: diatomic curves, energy conservation
- stability and reactivity: heating, compression, combustion
- distribution shifts: gas adsorption, vacancy migration
- thermodynamics: equation of state, phase transitions

💧 mlipbenchmarks [60, 104]

Molecules, 15 models: accuracy, MD speed and GPU memory; all ran stable MD, so accuracy against speed decides

📖 Before you commit to a model

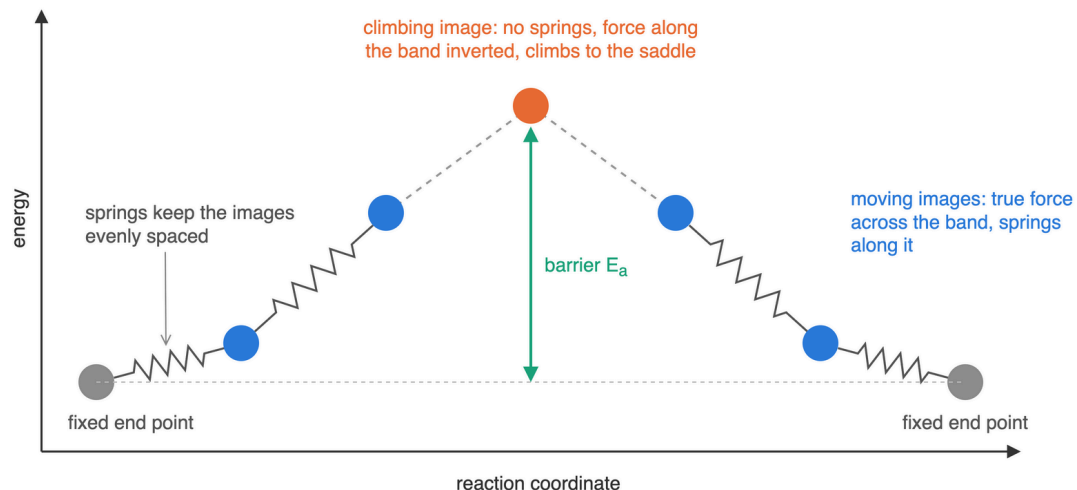
- Benchmark your own property class
- Run an ensemble: disagreement flags low confidence
- Watch stability, not only static errors



Heating MD: 120 NVT runs from 300 to 3000 K in 10 ps, one A100. Left: share of runs still valid; right: MD speed vs size. Chiang et al., [arXiv:2509.20630](https://arxiv.org/abs/2509.20630), Fig. 3(a) [61] (CC BY 4.0, cropped)

- Valid runs range from about 97% (ORBv2) to about 44% (M3GNet)
- Stability and speed rank models differently from static errors

Transition states and NEB with MLIPs



The barrier E_a sets reaction and diffusion rates, exponentially. NEB relaxes 5 to 9 images together.

With DFT

Forces on every image at every step: hundreds to thousands of DFT calls per path

With an MLIP

Same algorithm, milliseconds per force call: thousands of paths become practical

Nudged elastic band: images joined by springs; the climbing image finds the saddle [82].

CatTSunami, 932 surface reactions, models never trained on reactions [75]:

- Transition state within 0.1 eV of DFT in 91% of cases (largest model)
- With 2 DFT relaxations and 1 DFT single point: 88% at a 28x speed-up
- CO hydrogenation on Rh(111): 19,000 NEBs in 12 GPU days; DFT would have taken an estimated 52 GPU years

Zero-shot barriers: how good are they?

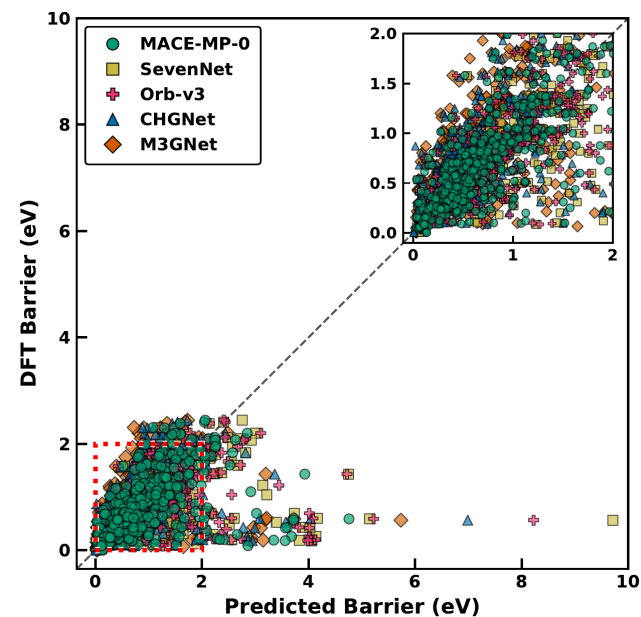
Battery ions: 574 paths, five models as provided [74] [107]

- Plain NEB against DFT-NEB (GGA)
- MAE 0.31 (MACE-MP-0) to 0.35 eV (M3GNet); 0.20 to 0.26 eV without each model's outliers above 1 eV
- Good or bad barrier at 0.5 eV: 74% (M3GNet) to 85% (Orb-v3) correct
- MLIP paths are a better DFT starting guess in about two thirds of cases

FPBench: 154 ion-migration paths, full CI-NEB [76] [108]

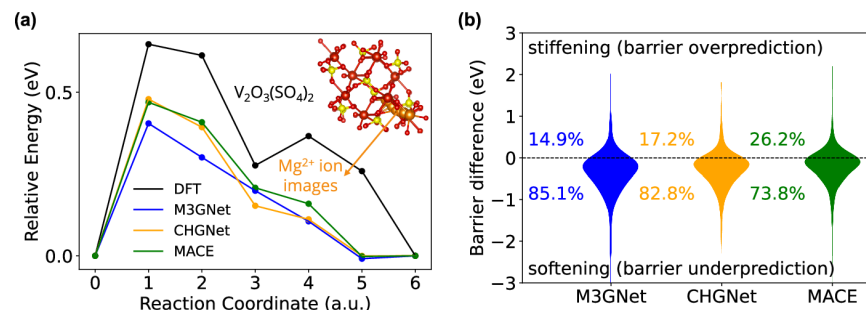
- MAE 0.05 eV (UMA), 0.06 eV (MACE-MPA-0), up to 0.17 eV for older models
- Most of the error sits in the energy surface, not in the NEB workflow

Newer, broader training data helps. DFT-NEB itself carries about 0.06 eV [74].



MLIP vs DFT-NEB barriers, 574 paths. Bheemaguli, Xiao & Sai Gautam, Fig. 2 [74] (CC BY 4.0, cropped)

MLIP barriers run low: DFT confirms the saddle



Mg^{2+} path in $\text{V}_2\text{O}_3(\text{SO}_4)_2$ and 477 barrier differences (MLIP minus DFT). Deng et al., npj Comput. Mater. 11, 9 (2025), Fig. 3 [69] (CC BY 4.0, unmodified)

- Trained mostly near equilibrium, universal MLIPs make the energy surface too flat, so barriers come out low [69]
- 470 Mg-ion paths: MAE 0.34 (MACE), 0.39 (CHGNet), 0.49 eV (M3GNet), shifted negative [69]. CHGNet and M3GNet under-estimate 73% and 78% of 574 paths [74]
- Fine-tuning on even one structure removes much of the bias [69]

Pattern: the MLIP finds the path, DFT confirms the saddle.

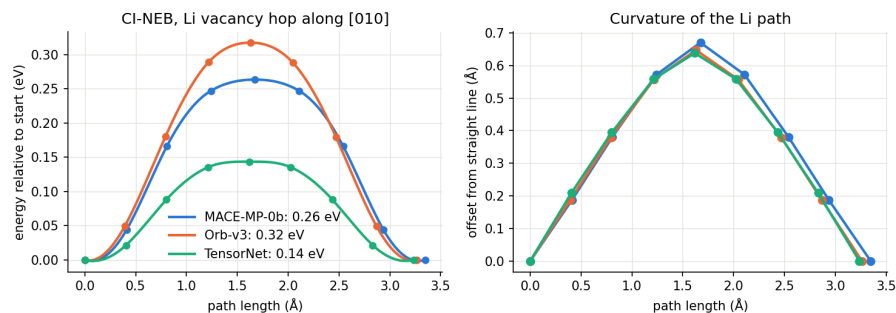
- CatTSunami: all MLIP 2200x faster, 70% success; add 2 DFT relaxations and 1 single point, 28x faster, 88% [75]
- Molecules: MACE-OMol25 path, then DFT: 96.6% success with 3.8 DFT gradients per reaction, 94 to 96% fewer than DFT alone [77]

Our LUMI run: a Li hop in LiFePO₄

Olivine cathode, 111 atoms, one Li vacancy, 7 images, CI-NEB, one MI250X GCD, float64, one run ([lesson A5](#)).

- All three models find the curved [010] path (0.64 to 0.67 Å off the straight line), as predicted [\[80\]](#) and seen by neutron diffraction [\[81\]](#)
- MACE and Orb land within 0.05 eV of DFT; TensorNet gives half, as softening predicts
- No spins or charges: the models cannot place the Fe³⁺ hole, which moves the DFT barrier by almost 0.2 eV [\[79\]](#)

Method	Barrier	Force calls	NEB time
MACE-MP-0b small	0.26 eV	343	12.0 s
Orb-v3 conservative	0.32 eV	329	19.4 s
TensorNet MatPES	0.14 eV	252	7.3 s
GGA DFT [78]	0.27 eV		
GGA+U DFT [79]	0.29 eV		



Left: CI-NEB energy profiles. Right: Li offset from the straight line. LUMI run.

Choosing an MLIP: rules of thumb

If your system is	Start with	Why
Periodic materials: crystals, alloys, oxides	• Orb-v3 first: fast, conservative for MD, D3 option [30, 31] • Also SevenNet-Omni [46], MatRIS [47] • On LUMI: NequIP-OAM [19] or MACE	• Top five within 0.012 CPS [51]: pick on speed, licence, GPU • Our Orb-v3 Li hop in LiFePO₄: within 0.05 eV of DFT (A5)
Molecules, liquids, electrolytes	• Hybrid-DFT models: UMA 'omol' [66], OrbMol • MACE-OFF [67], AIMNet2	• Electrolyte density R² 0.98 vs 0.34 to 0.45 for materials models [45]
Layered, surfaces, MOFs, adsorption	• PBE model + D3, added once • Or UMA 'oc20', 'odac', 'omc' heads	• Graphite: Orb-v3 4.3 Å, +D3 3.44 Å, exp. 3.34 Å (A2; [68])
Phonons, elastic, defects, barriers	• Rank by κSRME, not F1 (NequIP-OAM-XL, PET-OAM-XL) • Plan to fine-tune	• Universal models soften the PES [69] • Mo C₁₁: 46% → 3% error once fine-tuned [54]
Amorphous or out of distribution	• Largest broad-data models (UMA-M-1p1 was best) • Compare several; fine-tune	• AM26: of 41 models only UMA-M-1p1 stayed below 15% on all five systems [55]
Long MD, NPT, diffusion	• Conservative forces • Dyna-Mat Pareto models: orb-v2, GRACE-2L-OAM, MACE-MH-1-OMAT	• Lowest-force-error tier had the worst pressure, 3.40 GPa, mostly UMA on one alloy [59]

Then check before you trust it (next slide).

Before you trust a model: five checks

1 · Model card

Training set, DFT functional, elements covered, task head

2 · Right reference

Compare with its own DFT functional, not directly with experiment

3 · Your property

Test your property class, not the headline score

4 · Two or three models

Run them side by side; disagreement is a warning

5 · Pilot run

A short MD or relaxation first; fine-tune if it is off

One size does not fit all

MatRIS beats NequIP-OAM-XL on stability F1 (0.921 vs 0.906) but not on thermal-conductivity error (0.22 vs 0.13). “Different applications will have different requirements” [\[60\]](#).

6 · Outlook, then hands-on

Beyond energies and forces: long-range physics

The gap

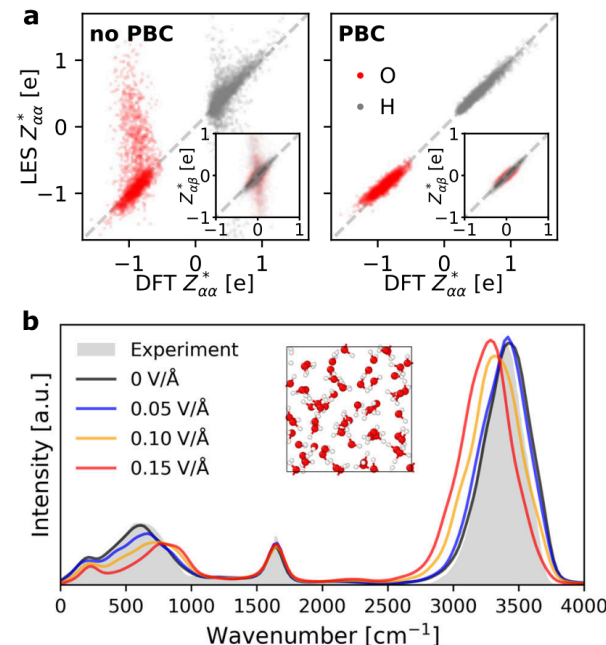
Standard MLIPs see only neighbours within a cutoff, so they miss long-range electrostatics: ions, interfaces, polar materials

The idea

Learn hidden (latent) charges from energies and forces alone, then add the long-range part with an Ewald sum; no charge training data needed

What it unlocks

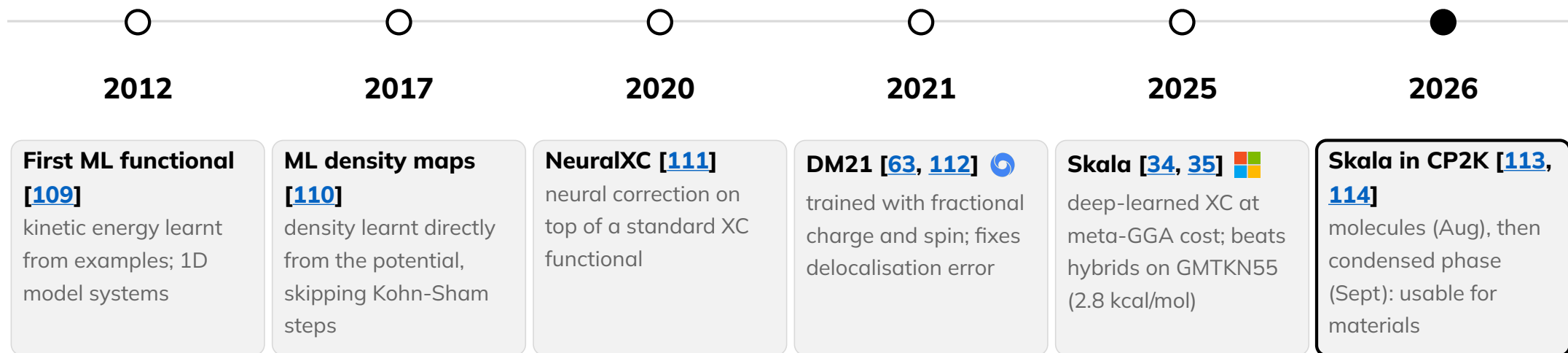
- Born effective charges and polarisation (panel a)
- Infrared spectra, also under an applied field (panel b)
- Ferroelectrics such as PbTiO_3 , and ionic conduction in superionic water



Liquid water. (a) Born effective charges, model vs DFT; (b) infrared spectrum vs experiment (grey) and under a field. Zhong, Kim, King & Cheng, npj Comput. Mater. 11, 384 (2025), Fig. 1 [62] (CC BY 4.0, cropped)

Machine learning inside DFT itself

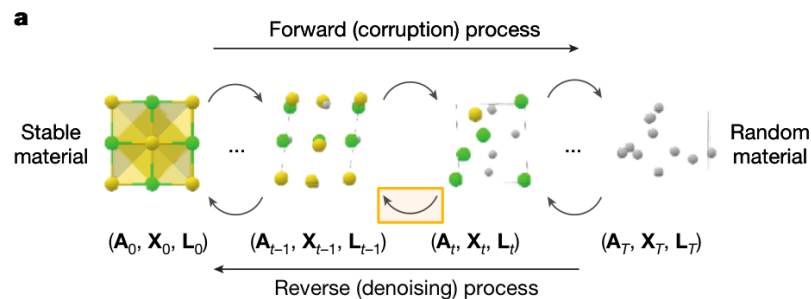
DFT is exact except for one term, the exchange-correlation (XC) functional, which must be approximated: the rungs of Jacob's ladder. Machine learning can learn that term from accurate data.



- Skala today: open code (microsoft-skala) with PySCF, GPU4PySCF and ASE interfaces, and now in CP2K for molecular and condensed-phase calculations
- Why it matters here: better, cheaper reference data for the next generation of MLIPs; the two directions reinforce each other

Generative models propose, MLIPs screen

A growing loop: generative models invent candidate crystals, and several MLIPs together relax and score them for stability.



Diffusion: a crystal is turned into noise, and the model learns the reverse. Zeni et al., Nature 2025, Fig. 1a [64] (CC BY 4.0, cropped)

MatterGen [64, 115]

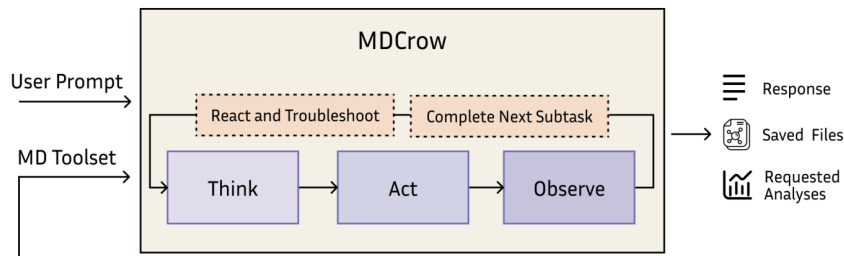
- Can be steered towards a target chemistry, symmetry or property
- More than twice as likely as earlier generators to give stable, unique and new crystals
- One candidate was made in the lab; a 2026 study argues it was already known [116]

LeMat-GenBench [64, 117]

- Scores 12 generators with an MLIP ensemble (MACE-MP, UMA, Orb)
- More stable output tends to mean less novel output; no model wins everywhere
- Synthesis and experimental checks remain the real bottleneck

Early agentic workflows

Language-model agents turn a plain-language goal into simulation steps: plan, call a tool, read the result, repeat.



Campbell et al., [arXiv:2502.09565](https://arxiv.org/abs/2502.09565), Fig. 1A [118] (CC BY 4.0, cropped)



MDCrow [118]

- 40+ tools; MD through OpenMM
- 25 tasks: 72% done correctly (gpt-4o) vs 28% for a bare LLM



El Agente Q [119]

- Quantum chemistry: xTB, ORCA, jobs via SLURM
- 6 exercises, about 88% average success



NVIDIA test, Aug 2026 [65]

- A coding agent wrote 45 batched GPU MACE pipelines
- None questioned an ill-posed task; an unstated thermostat damped Li diffusion 3 to 5x

Agents can already run the codes, but they do not yet question the physics: expert judgement stays in the loop.

Getting started



the lesson



Models and code



- MACE [27], SevenNet [46], Orb [31], MatterSim [48], FairChem/UMA [29], MatGL [23], DeePMD-kit [15]
- Engines: `ts`, TorchSim [57], kUPS [97], ALCHEMI [99]

Datasets

- MPtrj [24]: 1.6M configurations, the classic start
- OMat24 [41] and OMol25 [42]: 100M+ each, materials and molecules
- Alexandria: large open DFT database of crystals

Benchmarks

- Matbench Discovery [51]: stability, κ SRME, MD task [105]
- MLIP Arena [106]: physical tasks, stability
- mlipbenchmarks [104]: molecules, speed, memory

Help from ENCCS and Sweden AI Factory



- Lessons and workshops: enccs.se
- Access to LUMI, Leonardo and Arrhenius
- Compute and AI expertise: swedenaifactory.se
- training@enccs.se

Logos from each project's GitHub organisation, used to identify the software.

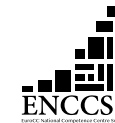
Summary

- ⊗ Universal MLIPs give near-DFT accuracy at a fraction of the cost, across the periodic table; they inherit the accuracy of their training data
- ⚙️ Pre-train, then fine-tune: it usually needs less data; zero-shot models can be off on mechanical properties and give barriers that are too low
- ☰ Choose by task: Orb-v3 and other OAM models for periodic materials, hybrid-DFT models for molecules; then run the five checks
- 🖥️ GPU engines (TorchSim, kUPS, ALCHEMI) and EuroHPC systems (LUMI, Leonardo, Arrhenius) make them practical at scale
- 📊 Trust deliberately: stability and your own property, not one error number; the leaderboard moves within months
- 🔭 The frontier is widening: long-range physics, ML inside DFT, generative discovery, early agents

“A poorly posed initial question results in AI scientific slop.” Shyue Ping Ong, Sept 2026 (materialize.ai [\[70\]](#))

Try a foundation MLIP on your own system, this week.

Hands-on: try it yourself



scan for the lesson

Everything in this talk, as a self-contained lesson: enccs.github.io/mlip-hands-on

Part	What you do	Saved result it shows
Background	The concepts of this talk, in one page	Potentials, foundation models, batching
Part A1	Batched relaxation with TorchSim	Leonardo A100: 5.05x for 64 structures (f64), 7.27x for 128 (f32); LUMI MI250X check
Part A2	Orb-v3 and MatGL TensorNet; D3 dispersion on graphite	Laptop CPU: layer spacing 3.44 Å vs 3.34 Å in experiment
Part A3	MatGL on LUMI: CsCl relax and MD, 23-crystal lattice benchmark, fine-tune vs scratch	Lattice MAPE 1.11%; fine-tuning about 5x lower energy error
Part A4	Fine-tune TensorNet from PBE to r2SCAN on LUMI	84 Li structures: 60 vs 513 meV/atom, forces 128 vs 427 meV/Å
Part A5	CI-NEB of a Li hop in LiFePO ₄ with MACE, Orb and TensorNet	LUMI MI250X: 0.26, 0.32, 0.14 eV vs 0.27 to 0.29 eV DFT
Part A6	Orb and UMA: Si, Cu crystals; O ₂ and water with charge and spin	LUMI MI250X: lattice within 0.5% of PBE; water IE 12.64 eV (UMA), 6.07 eV (OrbMol) vs 12.62 eV
Part B (B1-B8)	MD with ALCHEMI and LAMMPS	GH200 (Arrhenius, JUPITER): one trajectory to many GPUs
Choosing an MLIP	Rules of thumb and checks, with sources	The rules-of-thumb slide, in full
Further reading	Where to go next	Eight reviews, with links

Results shown are saved runs (Leonardo A100, LUMI MI250X, GH200, laptop CPU for graphite); Leonardo is in maintenance today.

Thank you

Thank you



Universal Machine Learning Interatomic Potentials on HPC, a joint webinar from ENCCS and Sweden AI Factory

- Hands-on lesson: enccs.github.io/mlip-hands-on (slides shared after the webinar)
- Contact and access support: training@enccs.se
- Compute and AI support from Sweden AI Factory: swedenaifactory.se
- Upcoming events and lessons: enccs.se/events · enccs.github.io/lessons



Hands-on lesson

Questions?



**Co-funded by
the European Union**



ENCCS is the Swedish node of the EuroCC 3 project. EuroCC 3 has received funding from the European High-Performance Computing Joint Undertaking (JU) under Grant Agreement No. 101306701. The JU receives support from the European Union's Digital Europe Programme and Germany, Albania, Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Montenegro, the Netherlands, North Macedonia, Norway, Poland, Portugal, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Türkiye, and Kosovo. The project is supported by the European High-Performance Computing Joint Undertaking and its members.

Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the granting authority (EuroHPC Joint Undertaking). Neither the European Union nor the granting authority can be held responsible for them.

ENCCS has also received national funding through Vinnova and the Swedish Research Council (VR).

Further reading: reviews

Start with the first two, both 2026 overviews.

- Medford & Sholl. Replacing quantum chemistry with MLIPs: revolution or evolution? ACS Cent. Sci. 12, 891 (2026). [doi:10.1021/acscentsci.6c00615](https://doi.org/10.1021/acscentsci.6c00615)
- Yuan, ..., Head-Gordon. Foundation models for atomistic simulation of chemistry and materials. Nat. Rev. Chem. 10, 212 (2026). [doi:10.1038/s41570-025-00793-5](https://doi.org/10.1038/s41570-025-00793-5)
- Jacobs, Morgan et al. A practical guide to MLIPs: status and future. Curr. Opin. Solid State Mater. Sci. 35, 101214 (2025). [doi:10.1016/j.cossms.2025.101214](https://doi.org/10.1016/j.cossms.2025.101214)
- Liu et al. Fine-tuning universal MLIPs: a tutorial. J. Appl. Phys. 139, 041101 (2026). [doi:10.1063/5.0299305](https://doi.org/10.1063/5.0299305)
- Kim et al. Reactive MLIPs for chemistry and materials science. Chem. Rev. 126, 4467 (2026). [doi:10.1021/acs.chemrev.5c00728](https://doi.org/10.1021/acs.chemrev.5c00728)
- Behler. Four generations of high-dimensional neural network potentials. Chem. Rev. 121, 10037 (2021). [doi:10.1021/acs.chemrev.0c00868](https://doi.org/10.1021/acs.chemrev.0c00868)
- Unke et al. Machine learning force fields. Chem. Rev. 121, 10142 (2021). [doi:10.1021/acs.chemrev.0c01111](https://doi.org/10.1021/acs.chemrev.0c01111)
- Deringer, Caro & Csányi. MLIPs as emerging tools for materials science. Adv. Mater. 31, 1902765 (2019). [doi:10.1002/adma.201902765](https://doi.org/10.1002/adma.201902765)

References (1/5)

1. Behler & Parrinello. Generalized neural-network representation of high-dimensional potential-energy surfaces. Phys. Rev. Lett. 98, 146401 (2007). [doi:10.1103/PhysRevLett.98.146401](https://doi.org/10.1103/PhysRevLett.98.146401)
2. Code. github.com/CompPhysVienna/n2p2
3. Bartók et al. Gaussian Approximation Potentials: The Accuracy of Quantum Mechanics, without the Electrons. Physical Review Letters 104, 136403 (2010). [doi:10.1103/physrevlett.104.136403](https://doi.org/10.1103/physrevlett.104.136403)
4. Code. github.com/libAtoms/QUIP
5. Thompson et al. Spectral neighbor analysis method for automated generation of quantum-accurate interatomic potentials. Journal of Computational Physics 285, 316 (2015). [doi:10.1016/j.jcp.2014.12.018](https://doi.org/10.1016/j.jcp.2014.12.018)
6. Code. github.com/FitSNAP/FitSNAP
7. Shapeev. Moment Tensor Potentials: A Class of Systematically Improvable Interatomic Potentials. Multiscale Modeling & Simulation 14, 1153 (2016). [doi:10.1137/15m1054183](https://doi.org/10.1137/15m1054183)
8. Code. gitlab.com/ashapeev/mlip-2
9. Smith et al. ANI-1: an extensible neural network potential with DFT accuracy at force field computational cost. Chemical Science 8, 3192 (2017). [doi:10.1039/c6sc05720a](https://doi.org/10.1039/c6sc05720a)
10. Code. github.com/aigm/torchani
11. Smith et al. Approaching coupled cluster accuracy with a general-purpose neural network potential through transfer learning. Nat. Commun. 10, 2903 (2019). [doi:10.1038/s41467-019-10827-4](https://doi.org/10.1038/s41467-019-10827-4)
12. Schütt et al. SchNet – A deep learning architecture for molecules and materials. The Journal of Chemical Physics 148, 241722 (2018). [doi:10.1063/1.5019779](https://doi.org/10.1063/1.5019779)
13. Code. github.com/atomistic-machine-learning/schnetpack
14. Zhang et al. Deep Potential Molecular Dynamics: A Scalable Model with the Accuracy of Quantum Mechanics. Physical Review Letters 120, 143001 (2018). [doi:10.1103/physrevlett.120.143001](https://doi.org/10.1103/physrevlett.120.143001)
15. Code. github.com/deepmodeling/deepmd-kit
16. Drautz. Atomic cluster expansion for accurate and transferable interatomic potentials. Physical Review B 99, 014104 (2019). [doi:10.1103/physrevb.99.014104](https://doi.org/10.1103/physrevb.99.014104)
17. Code. github.com/ICAMS/python-ace
18. Batzner et al. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. Nat. Commun. 13, 2453 (2022). [doi:10.1038/s41467-022-29939-5](https://doi.org/10.1038/s41467-022-29939-5)
19. Code. github.com/mir-group/nequip
20. Batatia et al. MACE: higher-order equivariant message passing. [arXiv:2206.07697](https://arxiv.org/abs/2206.07697) (2022).
21. Code. github.com/ACEsuit/mace
22. Chen & Ong. A universal graph deep learning interatomic potential for the periodic table. Nature Computational Science 2, 718 (2022). [doi:10.1038/s43588-022-00349-3](https://doi.org/10.1038/s43588-022-00349-3)
23. Code. github.com/materializeai/matgl
24. Deng et al. CHGNet as a pretrained universal neural network potential. Nat. Mach. Intell. 5, 1031 (2023). MPtrj dataset. [doi:10.1038/s42256-023-00716-3](https://doi.org/10.1038/s42256-023-00716-3)
25. Code. github.com/CederGroupHub/chgnet
26. Batatia et al. A foundation model for atomistic materials chemistry. [arXiv:2401.00096](https://arxiv.org/abs/2401.00096) (2024).
27. Code. github.com/ACEsuit/mace-foundations
28. Wood et al. UMA: a family of universal models for atoms. [arXiv:2506.23971](https://arxiv.org/abs/2506.23971) (2025).
29. Code. github.com/facebookresearch/fairchem
30. Rhodes et al. Orb-v3. [arXiv:2504.06231](https://arxiv.org/abs/2504.06231) (2025).
31. Code. github.com/orbital-materials/orb-models
32. Kavanagh, Tan, Wang et al. Fast and accurate equivariant foundation models for atomistic simulation (NequIP, Allegro). [arXiv:2607.28461](https://arxiv.org/abs/2607.28461) (2026).
33. Li et al. Universal Machine-learning Molecular Dynamics at the Speed of Empirical Potentials. [arXiv:2608.19041](https://arxiv.org/abs/2608.19041) (2026).
34. Luise et al. Skala, a learned exchange-correlation functional. [arXiv:2506.14665](https://arxiv.org/abs/2506.14665) (2025).
35. Code. github.com/microsoft/skala
36. Froitzheim, Müller, Hansen & Grimme. g-xTB. ChemRxiv (2025). [doi:10.26434/chemrxiv-2025-bjxvt](https://doi.org/10.26434/chemrxiv-2025-bjxvt)

References (2/5)

37. Pracht, Grimme, Bannwarth et al. CREST. J. Chem. Phys. 160, 114110 (2024). [doi:10.1063/5.0197592](https://doi.org/10.1063/5.0197592)
38. Xia, Thie, Soon & Barca. NN-xTB. Nat. Commun. 17, 7302 (2026). [doi:10.1038/s41467-026-73184-z](https://doi.org/10.1038/s41467-026-73184-z)
39. Jose & Walsh. GFN1-xTB-assisted ML for MOF screening. J. Chem. Theory Comput. 22, 8531 (2026). [doi:10.1021/acs.jctc.6c00979](https://doi.org/10.1021/acs.jctc.6c00979)
40. Friede, Hölzer, Ehlert & Grimme. dxtb. J. Chem. Phys. 161, 062501 (2024). [doi:10.1063/5.0216715](https://doi.org/10.1063/5.0216715)
41. Barroso-Luque et al. The Open Materials 2024 (OMat24) dataset and models. Nat. Comput. Sci. 6, 642 (2026). [doi:10.1038/s43588-026-00996-w](https://doi.org/10.1038/s43588-026-00996-w). [arXiv:2410.12771](https://arxiv.org/abs/2410.12771).
42. Levine et al. Open Molecules 2025 (OMol25). [arXiv:2505.08762](https://arxiv.org/abs/2505.08762).
43. Liao et al. EquiformerV2. [arXiv:2306.12059](https://arxiv.org/abs/2306.12059) (2023).
44. Yang et al. MatterSim: a deep-learning atomistic model across elements, temperatures and pressures. [arXiv:2405.04967](https://arxiv.org/abs/2405.04967) (2024).
45. Kumar et al. Prediction and Experimental Verification of Electrolyte Solvation Structure from an OMol25-Trained Interatomic Potential. [arXiv:2603.20183](https://arxiv.org/abs/2603.20183) (2026).
46. SevenNet-Omni. github.com/MDIL-SNU/SevenNet
47. MatRIS-10M-OAM. github.com/HPC-AI-Team/MatRIS
48. MatterSim v1.5M. github.com/microsoft/mattersim
49. Nequix. github.com/atomicarchitects/nequix
50. EquiformerV3-OAM. github.com/atomicarchitects/equiformer_v3
51. Matbench Discovery. matbench-discovery.materialsproject.org
52. Riebesell et al. A framework to evaluate ML crystal stability predictions (Matbench Discovery). Nat. Mach. Intell. 7, 836–847 (2025). [doi:10.1038/s42256-025-01055-1](https://doi.org/10.1038/s42256-025-01055-1)
53. Liu et al. A study on the fine-tuning performance of universal machine-learned interatomic potentials (Si, high-entropy alloy). [arXiv:2506.07401](https://arxiv.org/abs/2506.07401) (2025).
54. Liu et al. Fine-tuning universal MLIPs: a tutorial. J. Appl. Phys. 139, 041101 (2026). [doi:10.1063/5.0299305](https://doi.org/10.1063/5.0299305).
55. Fragapane & Deringer. Amorphous materials as a frontier challenge for universal interatomic potentials. [arXiv:2607.11384](https://arxiv.org/abs/2607.11384) (2026).
56. Engines: [TorchSim](https://github.com/ai-lab/TorchSim) (Radical AI), [kUPS](https://github.com/ai-lab/kUPS) (CuspAI), [ALCHEMI Toolkit](https://github.com/ai-lab/ALCHEMI-Toolkit) (NVIDIA).
57. github.com/TorchSim/torch-sim. github.com/TorchSim/torch-sim
58. docs.lumi-supercomputer.eu. docs.lumi-supercomputer.eu
59. Gawkowski et al. Dyna-Mat: End-to-end benchmarking of foundation machine learning interatomic potentials in finite-temperature ensembles. [arXiv:2607.03433](https://arxiv.org/abs/2607.03433) (2026).
60. Eastman, Pretti & Markland. Accuracy and efficiency benchmarks of pretrained MLIPs for molecular simulations. J. Chem. Theory Comput. 22, 6108 (2026). [doi:10.1021/acs.jctc.6c00130](https://doi.org/10.1021/acs.jctc.6c00130)
61. Chiang et al. MLIP Arena. NeurIPS 2025 Datasets & Benchmarks. [arXiv:2509.20630](https://arxiv.org/abs/2509.20630).
62. Zhong, Kim, King & Cheng. MLIPs can infer electrical response (Latent Ewald Summation). npj Comput. Mater. 11, 384 (2025). [doi:10.1038/s41524-025-01911-z](https://doi.org/10.1038/s41524-025-01911-z)
63. Kirkpatrick et al. Pushing the frontiers of density functionals (DM21). Science 374, 1385 (2021). [doi:10.1126/science.abj6511](https://doi.org/10.1126/science.abj6511)
64. Zeni et al. MatterGen: a generative model for inorganic materials design. Nature 639, 624–632 (2025). [doi:10.1038/s41586-025-08628-5](https://doi.org/10.1038/s41586-025-08628-5). Betala et al. LeMat-GenBench. [arXiv:2512.04562](https://arxiv.org/abs/2512.04562) (2025).
65. NVIDIA blog. developer.nvidia.com/blog/how-ai-coding-agents-can-unlock-materials-simulation-with-nvidia-alchemi-toolkit
66. UMA 'omol'. huggingface.co/facebook/UMA
67. MACE-OFF. github.com/ACEsuit/mace-off
68. Hazrati et al. Li intercalation in graphite: A van der Waals density-functional study. Physical Review B 90, 155448 (2014). [doi:10.1103/physrevb.90.155448](https://doi.org/10.1103/physrevb.90.155448)
69. Deng et al. Systematic softening in universal machine learning interatomic potentials. npj Computational Materials 11, 9 (2025). [doi:10.1038/s41524-024-01500-6](https://doi.org/10.1038/s41524-024-01500-6)
70. materialize.ai. www.materialize.ai/post/the-non-ai-pocalypse-in-materials-science
71. NAISS. Arrhenius technical description. www.naiss.se/resources/arrhenius-technical-description
72. Cheetham & Seshadri. Artificial intelligence driving materials discovery? Perspective on the article: Scaling deep learning for materials discovery. Chem. Mater. 36, 3490 (2024). [doi:10.1021/acs.chemmater.4c00643](https://doi.org/10.1021/acs.chemmater.4c00643)
73. Leeman et al. Challenges in high-throughput inorganic materials prediction and autonomous synthesis. PRX Energy 3, 011002 (2024). [doi:10.1103/PRXEnergy.3.011002](https://doi.org/10.1103/PRXEnergy.3.011002)

References (3/5)

74. Bheemaguli, Xiao & Sai Gautam. Evaluation of foundational machine learned interatomic potentials for migration barrier predictions. Digital Discovery 5, 1809 (2026). [doi:10.1039/D5DD000534E](https://doi.org/10.1039/D5DD000534E). [arXiv:2512.03642](https://arxiv.org/abs/2512.03642)
75. Wander, Shuaibi, Kitchin, Ulissi & Zitnick. CatTSunami: accelerating transition state energy calculations with pretrained graph neural networks. ACS Catal. 15, 5283 (2025). [doi:10.1021/acscatal.4c04272](https://doi.org/10.1021/acscatal.4c04272). [arXiv:2405.02078](https://arxiv.org/abs/2405.02078)
76. Amirian, Saravanan, Adams, Schwarz & Mo. FPBench: application-oriented error decomposition for foundation potentials. [arXiv:2609.05714](https://arxiv.org/abs/2609.05714) (2026).
77. Marks, Vandezande & Gomes. Reliable and efficient automated transition-state searches with machine-learned interatomic potentials. [arXiv:2604.00405](https://arxiv.org/abs/2604.00405) (2026).
78. Morgan, Van der Ven & Ceder. Li conductivity in Li_xMPO_4 (M = Mn, Fe, Co, Ni) olivine materials. Electrochem. Solid-State Lett. 7, A30 (2004). [doi:10.1149/1.1633511](https://doi.org/10.1149/1.1633511)
79. Dathar, Sheppard, Stevenson & Henkelman. Calculations of Li-ion diffusion in olivine phosphates. Chem. Mater. 23, 4032 (2011). [doi:10.1021/cm201604g](https://doi.org/10.1021/cm201604g)
80. Islam, Driscoll, Fisher & Slater. Atomic-scale investigation of defects, dopants, and lithium transport in the LiFePO_4 olivine-type battery material. Chem. Mater. 17, 5085 (2005). [doi:10.1021/cm050999v](https://doi.org/10.1021/cm050999v)
81. Nishimura et al. Experimental visualization of lithium diffusion in Li_xFePO_4 . Nat. Mater. 7, 707 (2008). [doi:10.1038/nmat2251](https://doi.org/10.1038/nmat2251)
82. Henkelman, Uberuaga & Jónsson. A climbing image nudged elastic band method for finding saddle points and minimum energy paths. J. Chem. Phys. 113, 9901 (2000). [doi:10.1063/1.1329672](https://doi.org/10.1063/1.1329672)
83. Fu et al. Learning smooth and expressive interatomic potentials for physical property prediction (eSEN). [arXiv:2502.12147](https://arxiv.org/abs/2502.12147) (2025).
84. Bigi et al. Pushing the limits of unconstrained machine-learned interatomic potentials (PET-OAM). [arXiv:2601.16195](https://arxiv.org/abs/2601.16195) (2026).
85. Bochkarev, Lysogorskiy & Drautz. Graph atomic cluster expansion. Phys. Rev. X 14, 021036 (2024). [doi:10.1103/PhysRevX.14.021036](https://doi.org/10.1103/PhysRevX.14.021036)
86. Code. github.com/lab-cosmo/upet
87. Code. github.com/ICAMS/grace-tensorpotential
88. Kim, You et al. Optimizing cross-domain transfer for universal machine learning interatomic potentials (SevenNet-Omni). [arXiv:2510.11241](https://arxiv.org/abs/2510.11241) (2025).
89. Code. github.com/mir-group/allegro
90. Zhou et al. MatRIS: toward reliable and efficient pretrained machine learning interatomic potentials. [arXiv:2603.02002](https://arxiv.org/abs/2603.02002) (2026).
91. Koker et al. Training a foundation model for materials on a budget (Nequix). [arXiv:2508.16067](https://arxiv.org/abs/2508.16067) (2025).
92. Liao et al. EquiformerV3: scaling efficient, expressive, and general SE(3)-equivariant graph attention transformers. [arXiv:2604.09130](https://arxiv.org/abs/2604.09130) (2026).
93. Kirkpatrick et al. Overcoming catastrophic forgetting in neural networks. PNAS 114, 3521 (2017). [doi:10.1073/pnas.1611835114](https://doi.org/10.1073/pnas.1611835114)
94. Kim, Lee, Oh et al. An efficient forgetting-aware fine-tuning framework for pretrained universal machine-learning interatomic potentials. npj Comput. Mater. 12, 26 (2026). [doi:10.1038/s41524-025-01895-w](https://doi.org/10.1038/s41524-025-01895-w)

References (4/5)

95. Cohen, Riebesell, Goodall et al. TorchSim: an efficient atomistic simulation engine in PyTorch. [arXiv:2508.06628](https://arxiv.org/abs/2508.06628) (2025).
96. TorchSim documentation. torchsim.github.io/torch-sim
97. Code. github.com/cusp-ai-oss/kups
98. CuspAI. kUPS: a molecular simulation engine for the AI era (blog, Apr 2026). medium.com/@CuspAI
99. Code. github.com/NVIDIA/nvalchemi-toolkit (v0.2.0); docs nvidia.github.io/nvalchemi-toolkit
100. ALCHEMI Toolkit-Ops, the GPU building blocks. github.com/NVIDIA/nvalchemi-toolkit-ops · docs nvidia.github.io/nvalchemi-toolkit-ops
101. NVIDIA blog. Building custom atomistic simulation workflows with NVIDIA ALCHEMI Toolkit. developer.nvidia.com/blog
102. NVIDIA blog. Accelerating AI-powered chemistry and materials science simulations with NVIDIA ALCHEMI Toolkit-Ops. developer.nvidia.com/blog
103. Fu et al. Forces are not enough: benchmark and critical evaluation for machine learning force fields with molecular simulations. TMLR (2023). [arXiv:2210.07237](https://arxiv.org/abs/2210.07237). Code: github.com/kyonofx/MDsim
104. Code. github.com/peastman/mlipbenchmarks
105. Matbench Discovery MD task. matbench-discovery.materialsproject.org/benchmarks/md
106. MLIP Arena code and leaderboard. github.com/atomind-ai/mlip-arena · huggingface.co/spaces/atomind/mlip-arena
107. Code and data. github.com/sai-mat-group/mlips-migration-barriers · Dataset-2: [doi:10.5281/zenodo.17483476](https://doi.org/10.5281/zenodo.17483476)
108. Code. github.com/mogroupumd/FPBench
109. Snyder, Rupp et al. Finding density functionals with machine learning. Phys. Rev. Lett. 108, 253002 (2012). [doi:10.1103/PhysRevLett.108.253002](https://doi.org/10.1103/PhysRevLett.108.253002)
110. Brockherde, Vogt et al. Bypassing the Kohn-Sham equations with machine learning. Nat. Commun. 8, 872 (2017). [doi:10.1038/s41467-017-00839-3](https://doi.org/10.1038/s41467-017-00839-3)
111. Dick & Fernandez-Serra. Machine learning accurate exchange and correlation functionals of the electronic density (NeuralXC). Nat. Commun. 11, 3509 (2020). [doi:10.1038/s41467-020-17265-7](https://doi.org/10.1038/s41467-020-17265-7) · Code: github.com/semodi/neuralxc
112. DM21 code. github.com/google-deepmind/deepmind-research
113. Pöschel, Pototschnig et al. Molecular implementation of the machine-learned Skala exchange-correlation functional in CP2K through GauXC. [arXiv:2608.19033](https://arxiv.org/abs/2608.19033) (2026).
114. Pototschnig, Pöschel et al. Native implementation of the machine-learned Skala exchange-correlation functional in CP2K: molecular and condensed-phase calculations. [arXiv:2609.34055](https://arxiv.org/abs/2609.34055) (2026).

References (5/5)

- 115. MatterGen code and weights. github.com/microsoft/mattergen · huggingface.co/microsoft/mattergen
- 116. Juelsholt. Continued challenges in high-throughput materials predictions: MatterGen predicts compounds from the training dataset. Mater. Horiz. 13, 5672 (2026). [doi:10.1039/D6MH00268D](https://doi.org/10.1039/D6MH00268D)
- 117. LeMat-GenBench code and leaderboard. github.com/LeMaterial/lemat-genbench · huggingface.co/spaces/LeMaterial/LeMat-GenBench
- 118. Campbell, Cox, Medina, Watterson & White. MDCrow: automating molecular dynamics workflows with large language models. [arXiv:2502.09565](https://arxiv.org/abs/2502.09565) (2025). Code: github.com/ur-whitelab/MDCrow
- 119. Zou et al. El Agente: an autonomous agent for quantum chemistry. Matter (2025). [doi:10.1016/j.matt.2025.102263](https://doi.org/10.1016/j.matt.2025.102263). [arXiv:2505.02484](https://arxiv.org/abs/2505.02484)